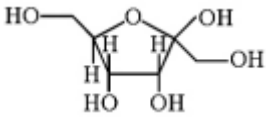
Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

Which descriptors fit the following sugar best?



(A) Ketose, furanose, α-configuration

(B) Ketose, furanose, β-configuration

(C) Aldose, pyranose, β-configuration

(D) Aldose, pyranose, α-configuration

Q.2

$$A \xrightarrow{4HIO_4} 3HCOOH + HCHO + OHC - COOH$$

What is the compound (A)

(A) Gluconic acid

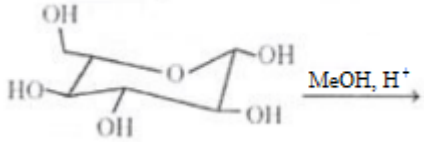
(B) Saccharic acid

(C) Glucitol

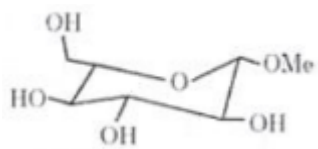
(D) Mannitol

Q.3

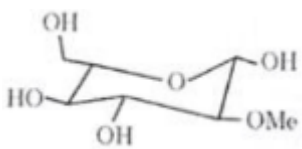
Predict the product



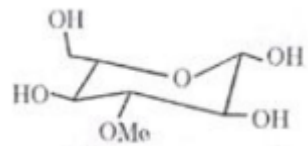
(A)



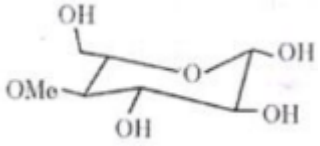
(B)



(C)

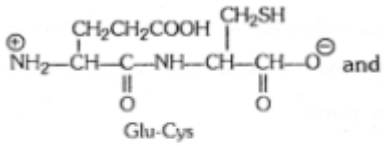


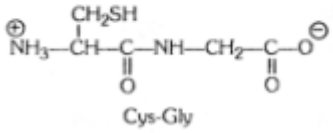
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(D) 

Q.4

A tripeptide (X) on partial hydrolysis gave two dipeptides Cys-Gly and Glu-Cys, i.e.





Identify tripeptide.

(A) Glu-Cys-Gly

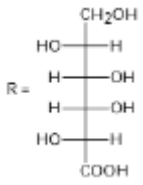
(B) Gty-Glu-Cys

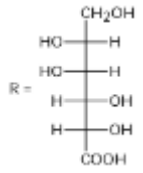
(C) Cys-Gly-Glu

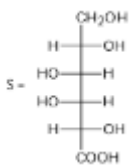
(D) Cus-Glu-Glu

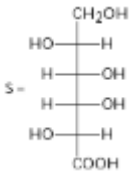
Q.5

Lactose on hydrolysis gives X and Y. Both X and Y are epimers but not anomers. Both X and Y on reaction with bromine water gives M and N respectively. Both X and Y on reaction with conc HNO_3 gives P and Q respectively. P is optically active and Q is optically inactive. R and S are enantiomers of M and N respectively. Then

(A) 

(B) 

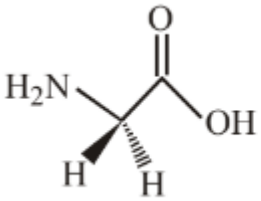
(C) 

(D) 

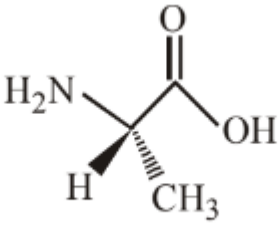
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Q.6

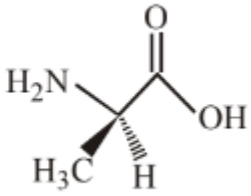
When an α-amino acid contains a substituent, there is possibility of optical isomers, for example, Lalanine and D-alanine are two enatiomers. What is the number of all possible linear tripeptides (repetrations of amino acids permitted) that can be formed from the following three amino acids : glycine, L-alanine and D-alanine as the starting materials in the condensation reaction?



Glycine (Gly)



L-alanine (L-Ala)



D-alamine (L-Ala)

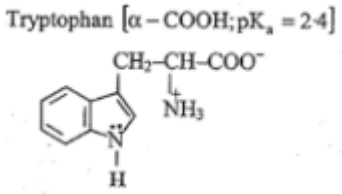
(A) $3C_2$

(B) 6

(C) 8

(D) 27

Q.7



does not migrate towards any pole, when it is placed in electrified solution of $pH5.9$. The pK_2 $\left(-\overset{+}{N}H_3\right)$ is

(A) 2.4

(B) 5.9

(C) $9 \cdot 4$

(D) 3.5

Q.8

Which of the following will act as inhibitor in free radical addition polymerisation?

(A) 

(B) CCl_4

(C) CBr_4

(D) $H - Br$

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Q.9

Which of the following statement is (are) correct -

(A) Glucose require phenyl hydrazine to give osazone

(B) Glyceraldehyde on oxidative cleavage with HIO_4 will give three different products

(C) Through Killiani-Fischer synthesis if we add two more C-atoms in chain to (D)-glyceraldehyde total number of possible compounds will be four.

(D) Hydrolysis of lactose give one unit of β-D-glucose and one unit of α-D-galactose

Q.10

Identify the correct statement (s) out of the following sugars -

(A) α-D-Glucopyranose has a hemi acetal structure and is a reducing sugar

(B) Methyl α-D-Glucopyranoside has acetal structure and is a non-reducing sugar

(C) α-D-Glucopyranose and β-D-Glucopyranose has anomers

(D) Lactose is a disaccharide and is a reducing sugar.

Q.11

Which of the following statements is/are correct -

(A) Glucose, mannose and fructose give the same osazone when treated with excess phenyl hydrazine

(B) Tyrosine is an amino acid containing a phenolic group

(C) Sucrose and fructose do not exhibit mutaroation

(D) The $C-O-C$ linkage that joins monosaccharide units in oligo and polysaccharides is known as glycosidic linkage

Q.12

Passage:
An α-amino acids A has molecular formula $C_3H_7NO_3$. A on treatment with methanol in presence of HCl yields $B\ [C_4H_{10}NO_3Cl]$ .B on further treatment with $PCl_5 yields' C'\ [C_4H_9NO_2Cl_2]$ which on acidic hydrolysis yields $D\ [C_3H_6NO_2Cl]$ .D on reduction with $Na(Hg)$ in dilute acidic medium yields alanine.
Question:
Compound A has how many functional groups ?

(A) 1

(B) 2

(C) 3

(D) 4

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Q.13

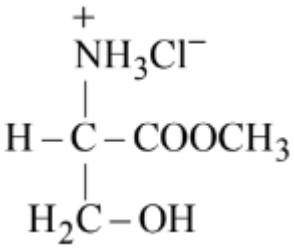
Passage:

An α-amino acids A has molecular formula $C_3H_7NO_3$. A on treatment with methanol in presence of HCl yields $B\ [C_4H_{10}NO_3Cl]$ .B on further treatment with PCl_5 yields $'C'\ [C_4H_9NO_2Cl_2]$ which on acidic hydrolysis yields $D\ [C_3H_6NO_2Cl]$ .D on reduction with $Na(Hg)$ in dilute acidic medium yields alanine.

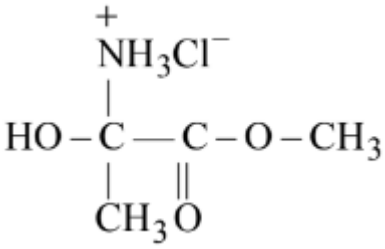
Question:

Compound B should be —

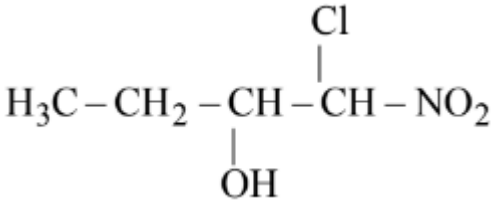
(A)



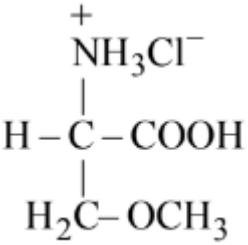
(B)



(C)



(D)



Q.14

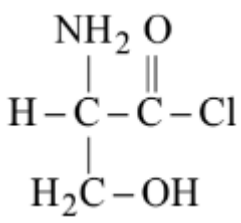
Passage:

An α-amino acids A has molecular formula $C_3H_7NO_3$. A on treatment with methanol in presence of HCl yields $B\ [C_4H_{10}NO_3Cl]$ .B on further treatment with PCl_5 yields $'C'\ [C_4H_9NO_2Cl_2]$ which on acidic hydrolysis yields $D\ [C_3H_6NO_2Cl]$ .D on reduction with $Na(Hg)$ in dilute acidic medium yields alanine.

Question:

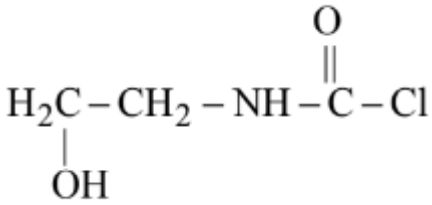
Compound D is —

(A)

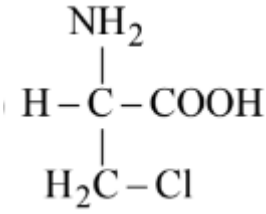


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(B)



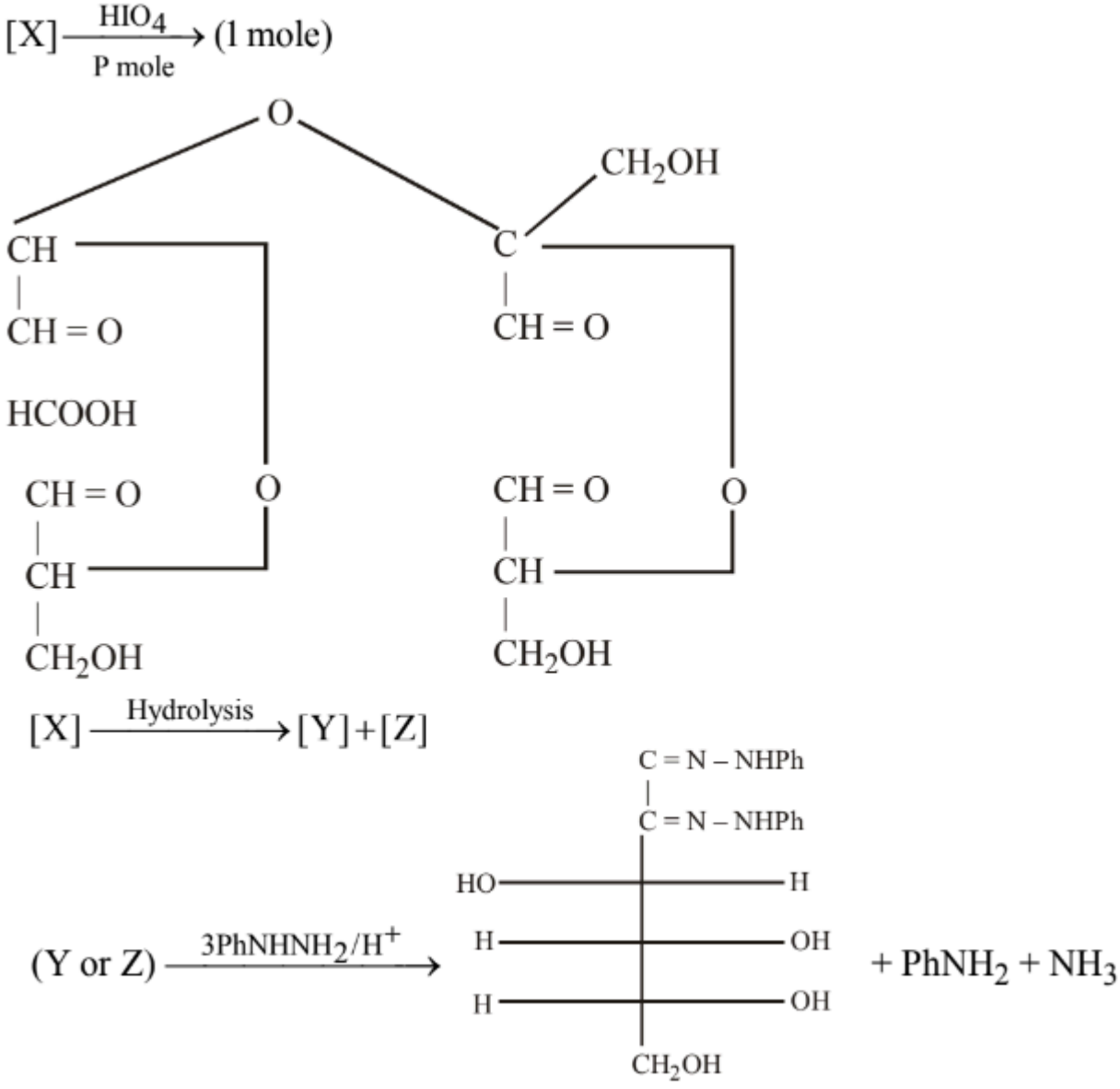
(C)



(D) (A) and (B) both are possible

Q.15

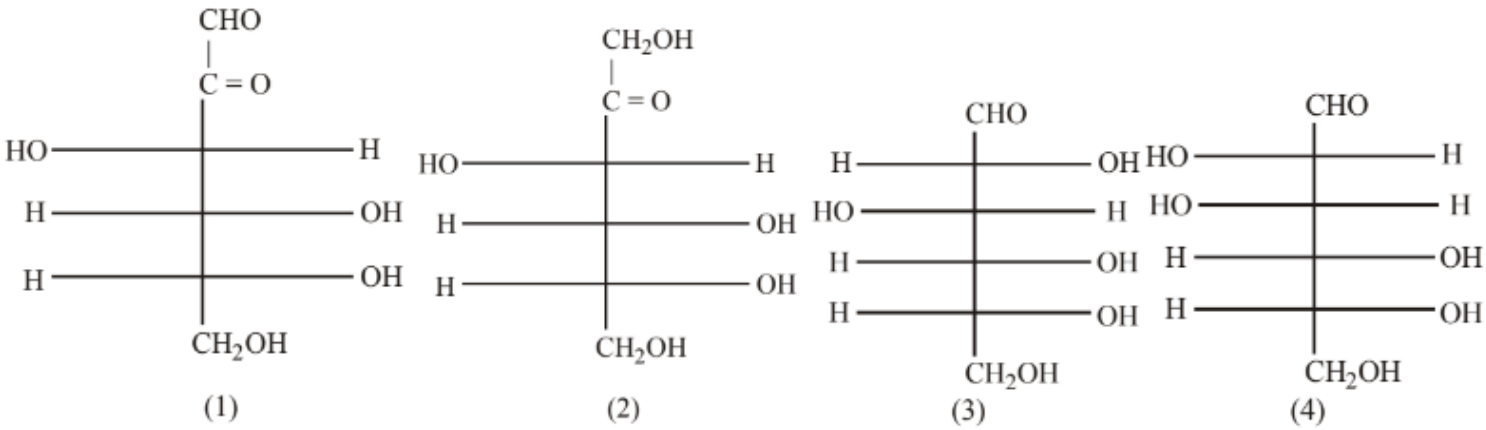
Passage:



Question:

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Compounds Y and Z can be –



(A) only 1

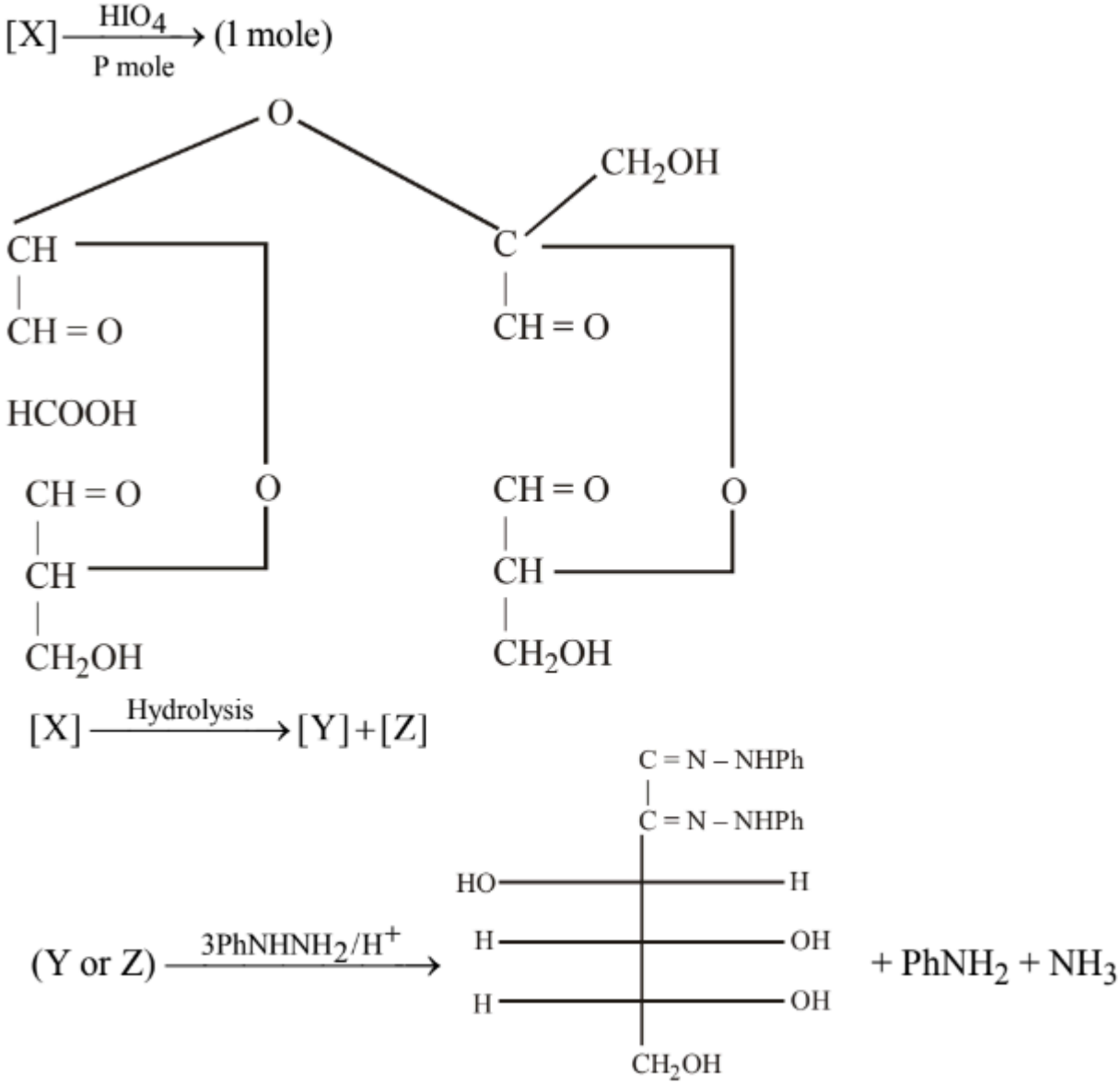
(B) 2,3

(C) 1,4

(D) $2, 3, 4$

Q.16

Passage:



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Question:

The value of P is —

(A) 2

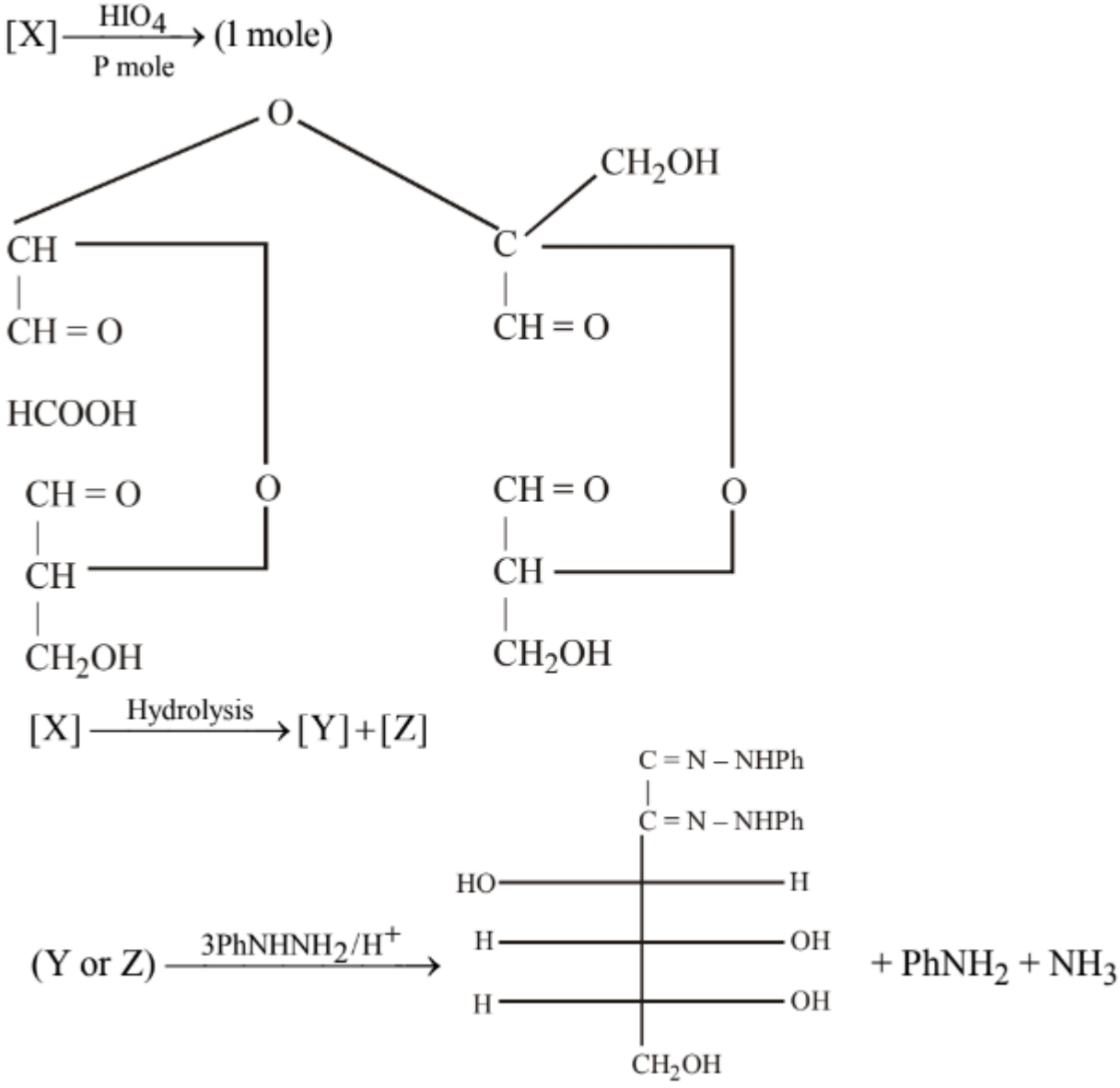
(B) 3

(C) 4

(D) 5

Q.17

Passage:



Question:

The true characteristics about X, Y and Z are —

(A) Oligosaccharide - X, Reducing - X, Monosaccaride - Y,Z, Mutarotation - Y,Z

(B) Oligosaccharide - X, Reducing - Y,Z, Monosaccaride - Y,Z, Mutarotation - Y,Z

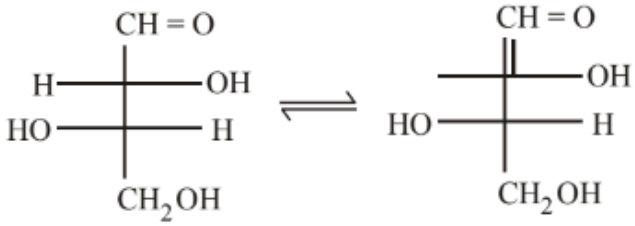
(C) Oligosaccharide - Y,Z, Reducing - Y,Z, Monosaccaride - X, Mutarotation - Y,Z

(D) Oligosaccharide - Y,Z, Reducing - X, Monosaccaride - X, Mutarotation - X

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Q.18

Match the following -

Column I	**Column II**
(A) α-D-glucopyranose ⇌ β-D-glucopyranose	(p) LDB Alberda van Ekenstein transformation
(B) Glucose ⇌ Mannose	(q) Mutarotation
(C) Fructose ⇌ Glucose	(r) Tautomerisation
(D) 	(s) Epimerisation

(A) (A) - p,q, (B) - r,p, (C) - p,s, (D) - r

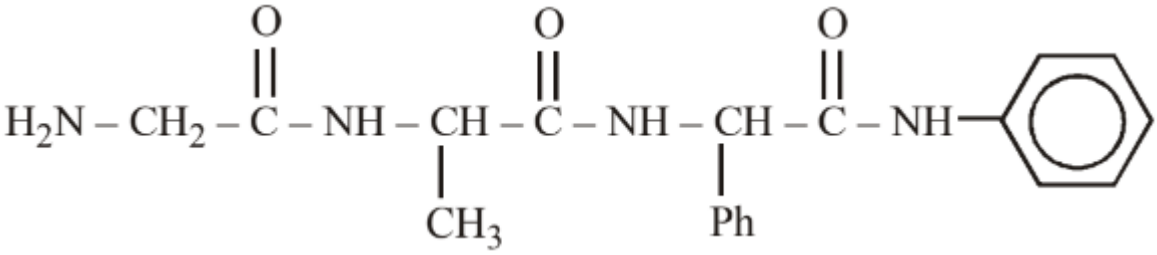
(B) (A) - q,r, (B) - p,s, (C) - p, (D) - p,r

(C) (A) - p,q, (B) - p,r, (C) - q,s, (D) - s

(D) (A) - s, (B) - p,r, (C) - r, (D) - q,s

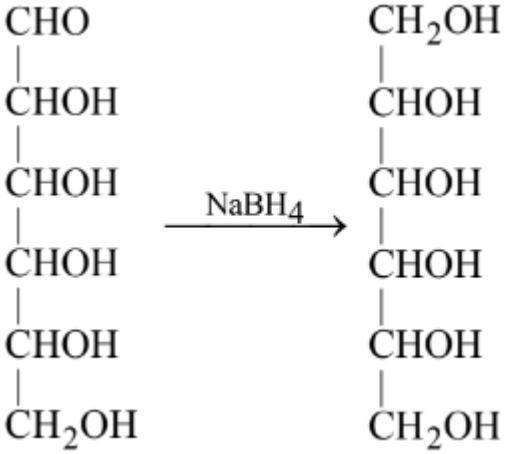
Q.19

The number of peptide bond(s) in the following molecule is/are



Q.20

Observe the following reaction and find out that how many number of reactant stereoisomers can be reduced to optically inactive meso products.



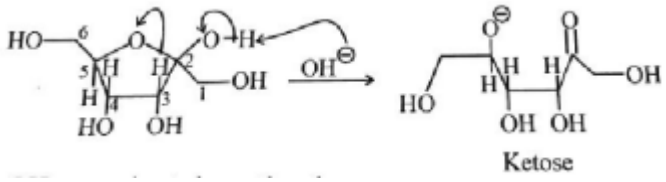
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Answers & Solutions

Q.1 Answer:

Ketose, furanose, β-configuration

Solution:



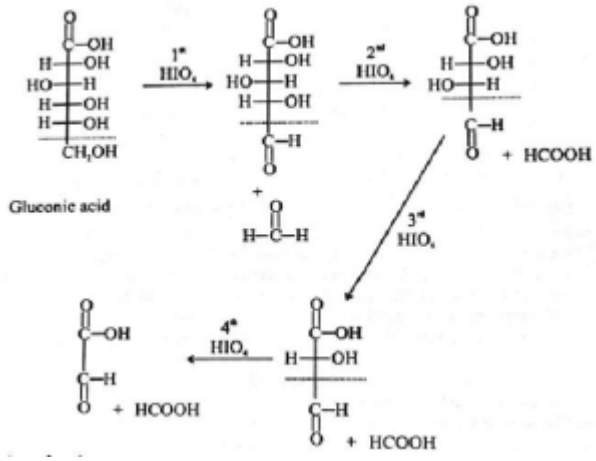
- OH group is at above the plane β-configuration

Five membered ring Furanose

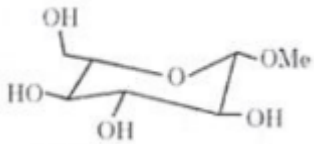
Q.2 Answer:

Gluconic acid

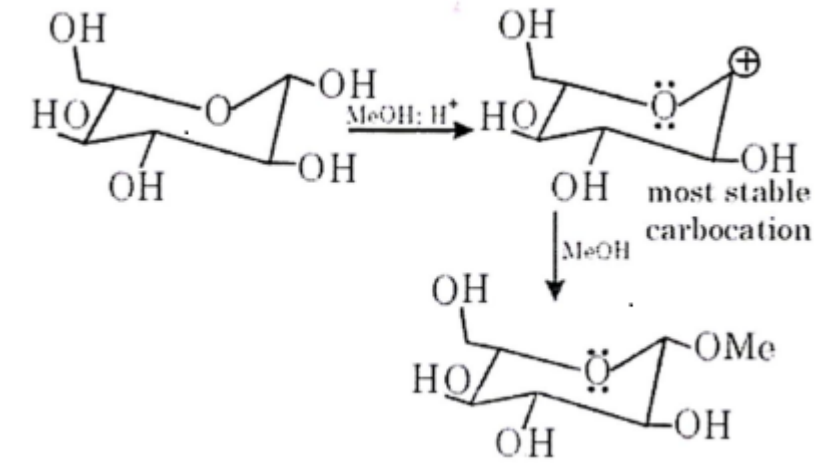
Solution:



Q.3 Answer:



Solution:



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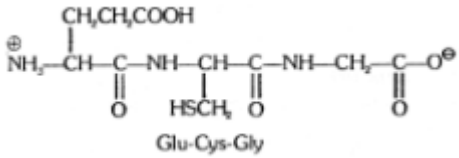
Q.4 Answer:

Glu-Cys-Gly

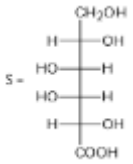
Solution:

Since the tripeptide on hydrolysis gave two dipeptides Glu-Cys and Cys-Gly,

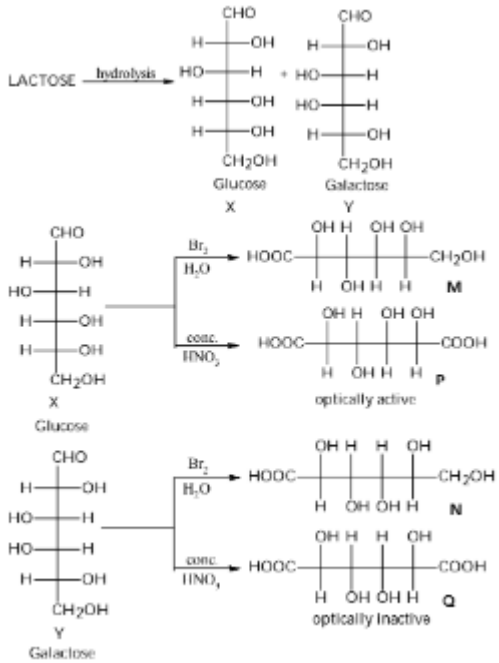
Hence cysteine must be in between glutamic acid and glycine as given below :



Q.5 Answer:



Solution:



Q.6 Answer:

27

Solution:

No. of all possible linear tripeptides $= 3^3 = 27$

Q.7 Answer:

$9 \cdot 4$

Solution:

$$\text{pH} = \frac{\text{pK}_{a_1} + \text{pK}_{a_2}}{2}$$
$$5.9 = \frac{2.4 + \text{pK}_{a_2}}{2}$$
$$\text{pK}_{a_2} = 9.4$$

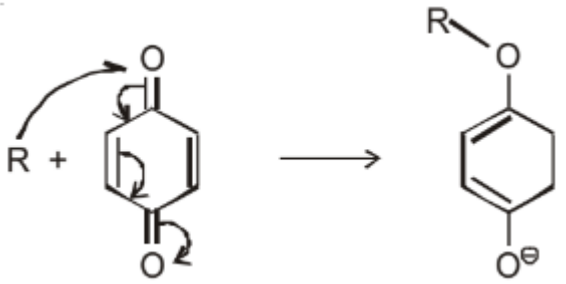
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Q.8 Answer:



Solution:

Such chain transfer agents which inhibit the process of polymerisation are called inhibitor. If the chain transfer agent forms a radical which is highly unreactive, the original chain gets terminated without new being initiated.



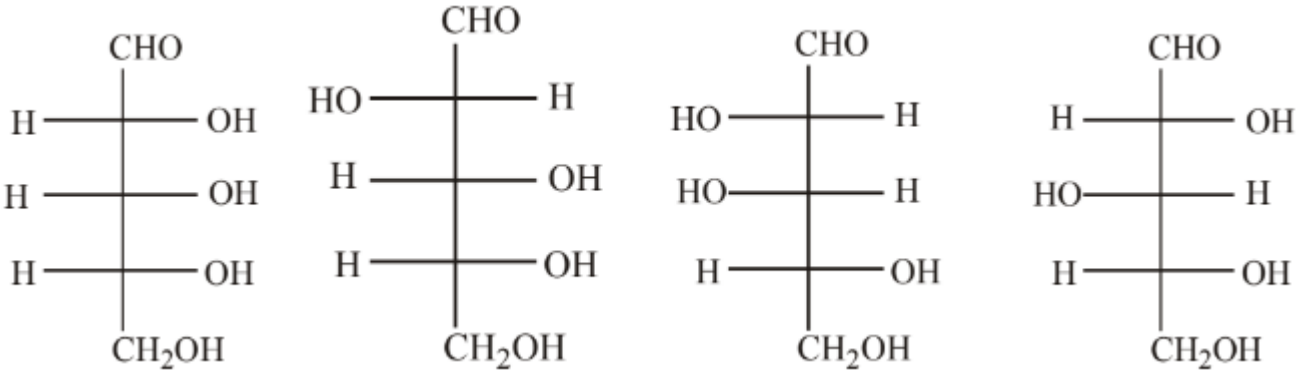
Q.9 Answer:

Glucose require phenyl hydrazine to give osazone

Solution:

Glucose requires 3 moles of phenyl hydrazine to give osazone.

Glyceraldehyde on cleavage with HIO_4 give 2 moles of $HCOOH + 1$ mole of $HCHO$ Through Killani-Fischer synthesis we will be able to get following four products.



Hydrolysis of lactose gives one unit of β-D-galactose and $\alpha - D - glu$cose .

Q.10 Answer:

α-D-Glucopyranose has a hemi acetal structure and is a reducing sugar

Solution:

Sugars that contain aldehyde groups that are oxidised to carboxylic acids are classified as reducing sugars.

Common test reagents are:

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1.Benedicts reagent $(CuSO_4/$ citrate $)$

2.Fehlings reagent $(CuSO_4/$ tartrate $)$

Reducing sugars reduce the Cu^{2+} to Cu^{+}which forms a red precipitate, copper (I) oxide.

In order for oxidation to occur, the cyclic form must first ring-open to give the reactive aldehyde

Any sugar that contains a hemi- acetal will be a reducing sugar.

Any Glycosides which are acetals are not reducing sugars.

(A) Lactose is a disaccharide sugar derived from galactose and glucose that is found in milk.it has only one of their two anomeric carbons involved in the

glycosidic bond, meaning that they can convert to an open-chain form with an aldehyde group.it is a reducing sugar.

(B) α-D glucopyranoside has acetal groups and hence is not a reducing sugar.

(C) Methyl- α-D-glucopyranoside has acetal groups and hence is not a reducing sugar.

(D) D-glucopyranose has hemiacetal structure and is reducing sugar as they can be changed to open chain form with an aldehyde group.

Q.11 Answer:

Glucose, mannose and fructose give the same osazone when treated with excess phenyl hydrazine

Solution:

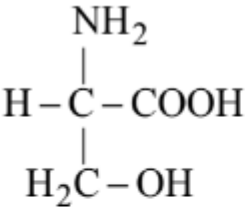
Sucrose does not show mutarotation but fructose and all other reducing sugars show the phenomenon of mutarotation.

Q.12 Answer:

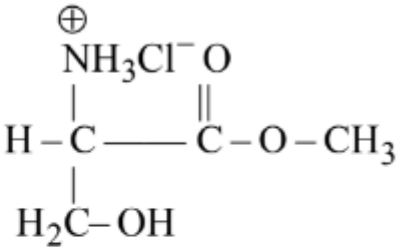
3

Solution:

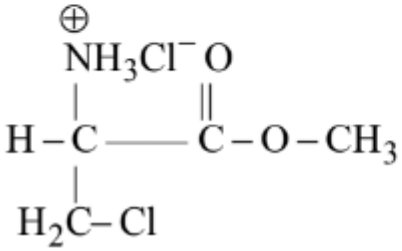
(A)



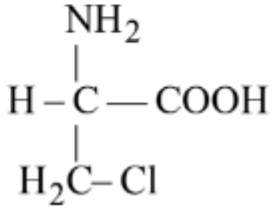
(B)



(C)

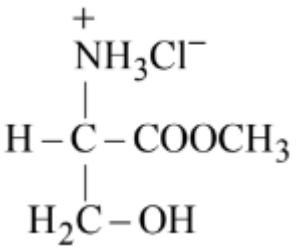


(D)

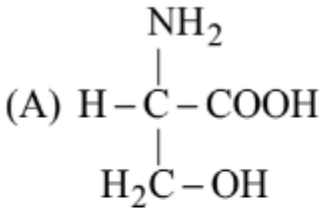


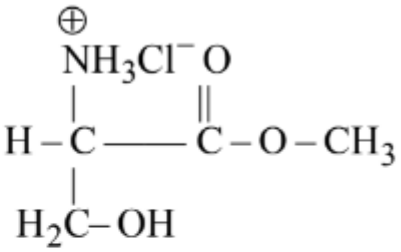
Q.13 Answer:

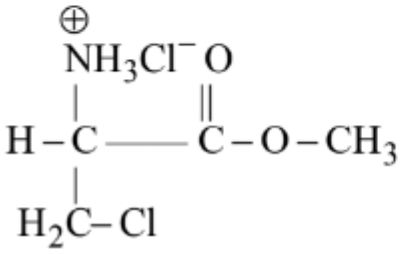
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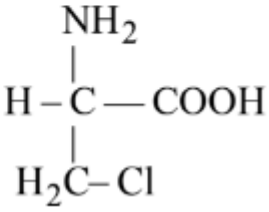


Solution:

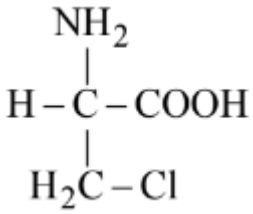
(A) 

(B) 

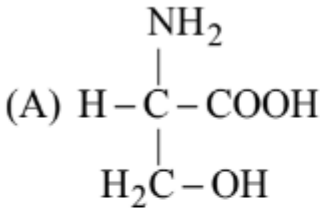
(C) 

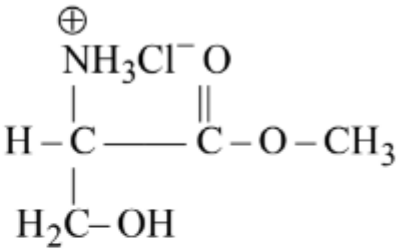
(D) 

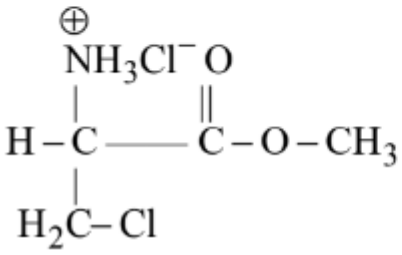
Q.14 Answer:

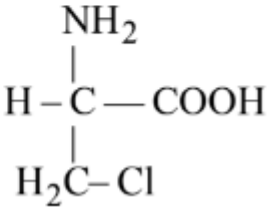


Solution:

(A) 

(B) 

(C) 

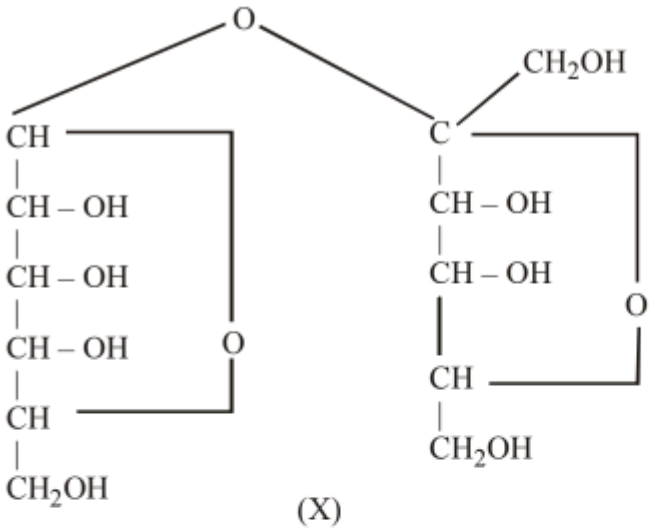
(D) 

Q.15 Answer:

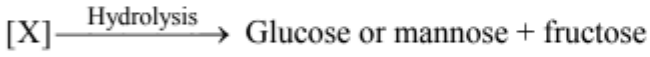
2, 3, 4

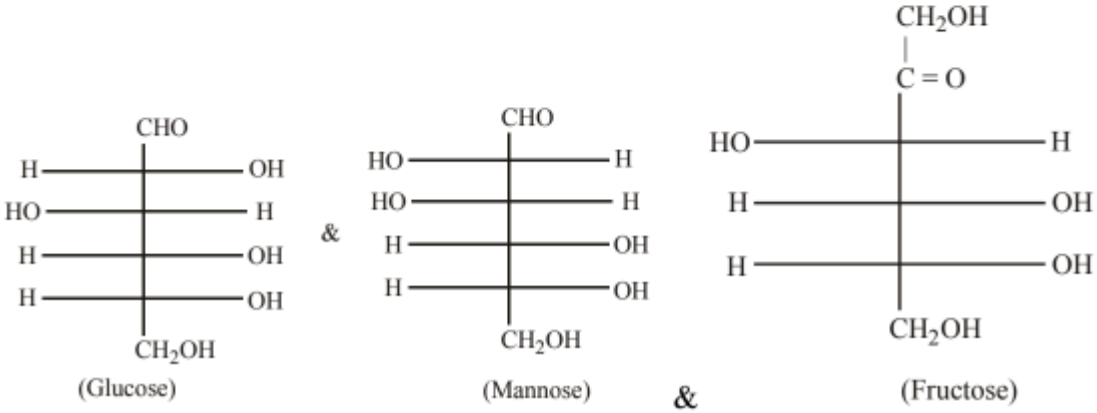
Solution:

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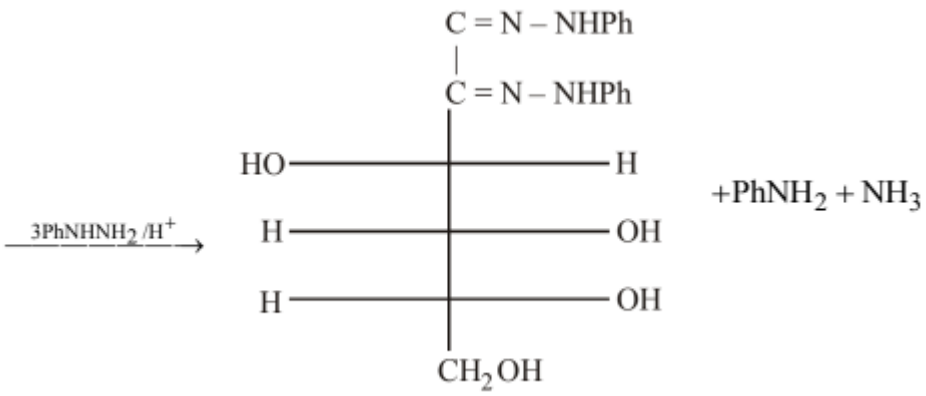


(X)





(Glucose) (Mannose) (Fructose)

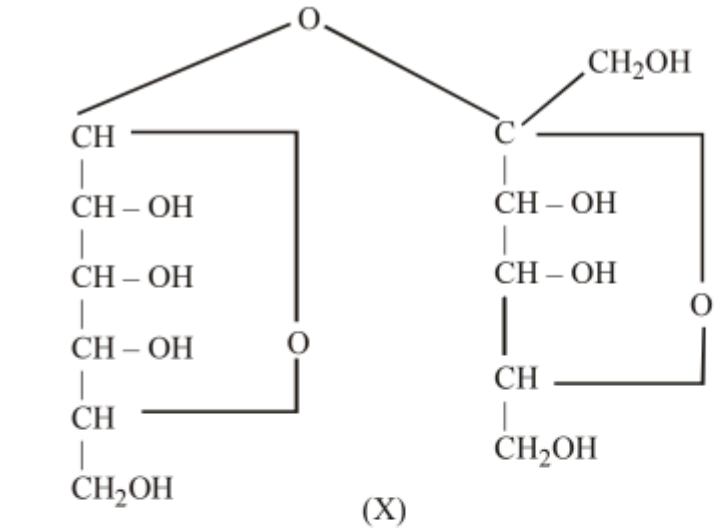


Q.16 Answer:

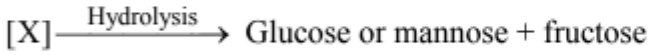
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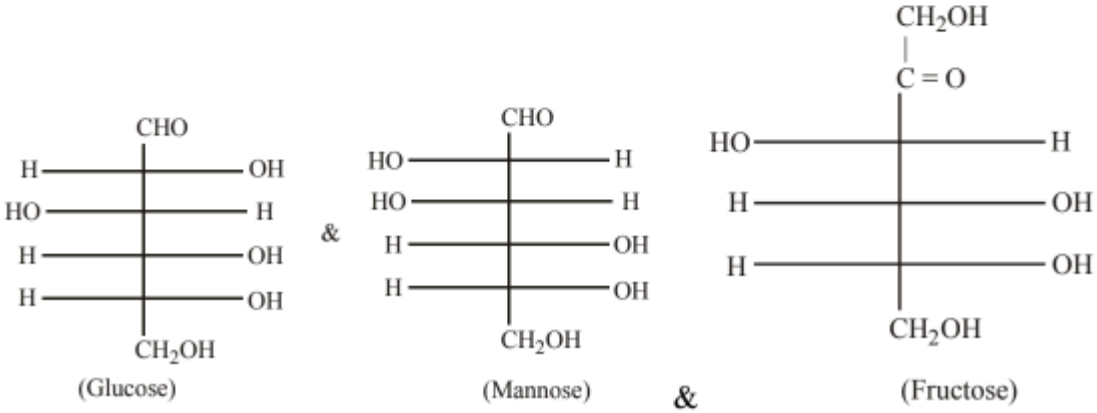
Solution:

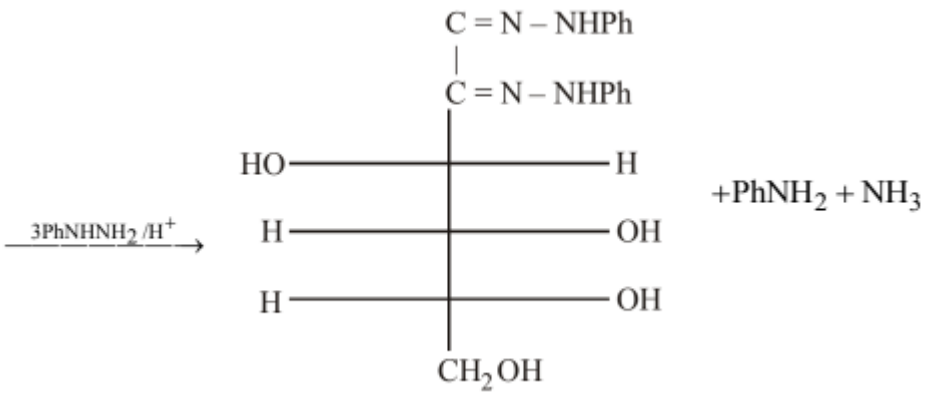
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(X)





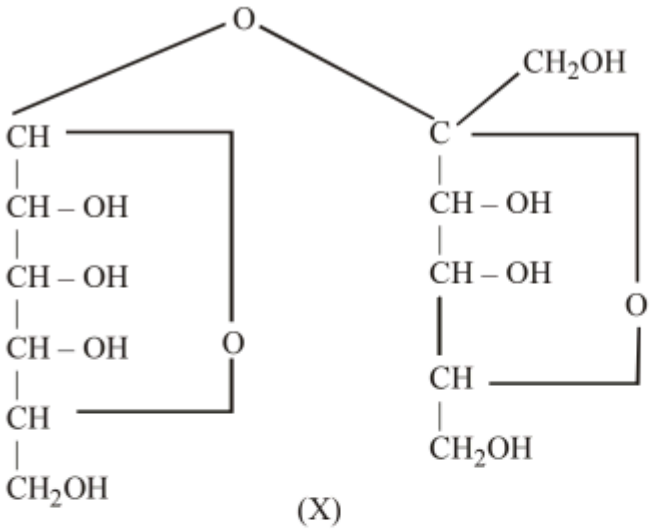


Q.17 Answer:

Oligosaccharide - X, Reducing - Y,Z, Monosaccaride - Y,Z, Mutarotation - Y,Z

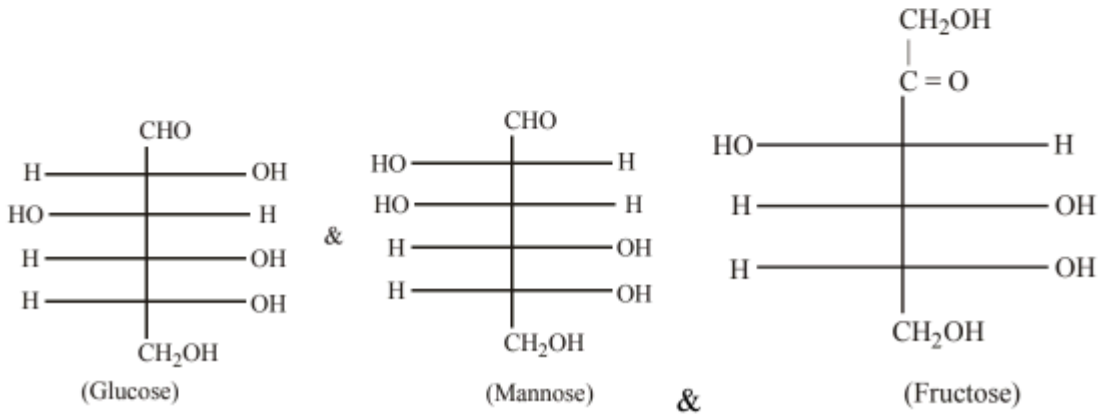
Solution:

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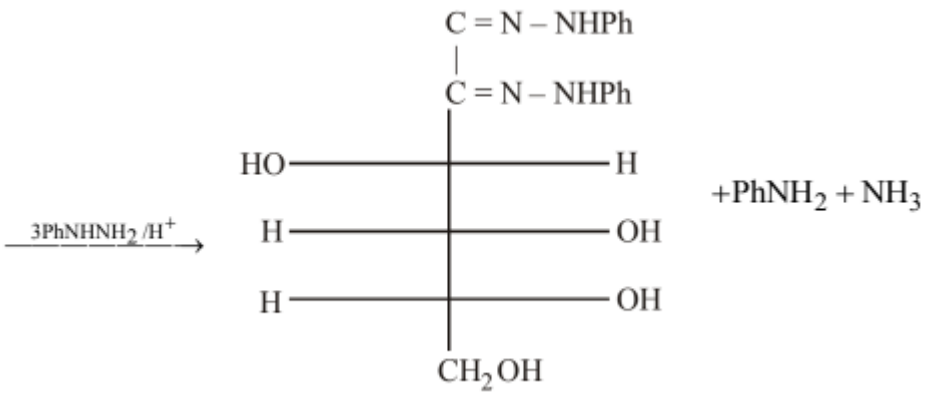


(X)

[X] $\xrightarrow{\text{Hydrolysis}}$ Glucose or mannose + fructose



(Glucose) & (Mannose) & (Fructose)



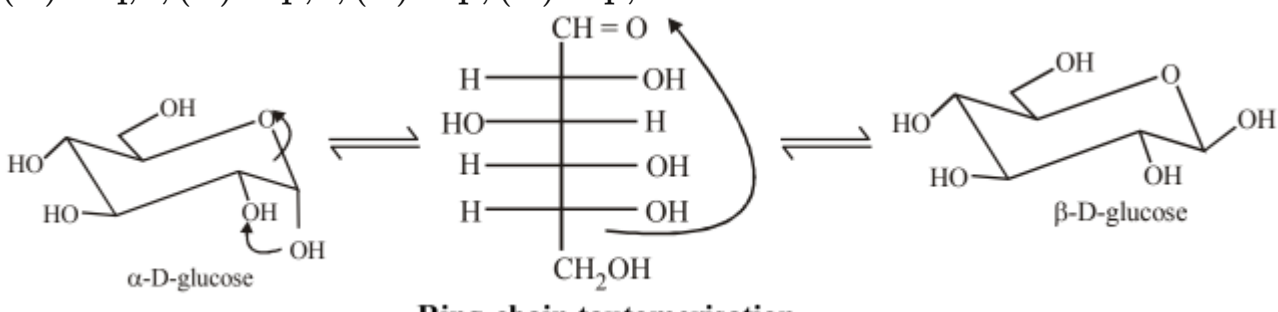
$+PhNH_2 + NH_3$

Q.18 Answer:

(A) - q,r, (B) - p,s, (C) - p, (D) - p,r

Solution:

$(A) - q, r, (B) - p, s, (C) - p, (D) - p, r$



α-D-glucose

Ring-chain tautomerisation

β-D-glucose

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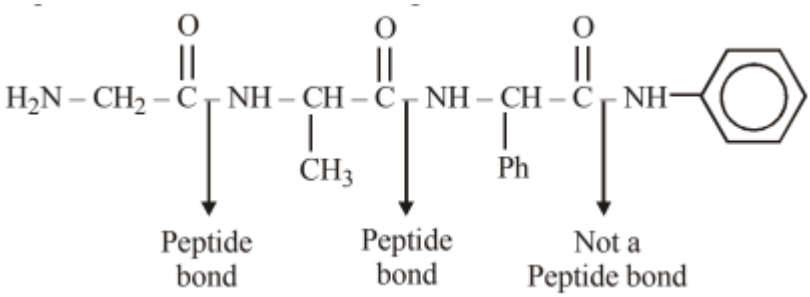
Hydroxy aldehyde and hemiacetal are tautomers.

Q.19 Answer:

—

Solution:

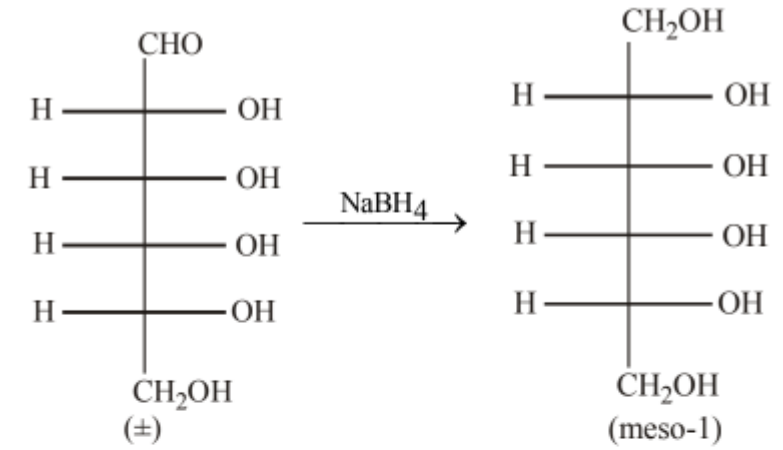
Peptide bond is the amide linkage between two α-amino acids.

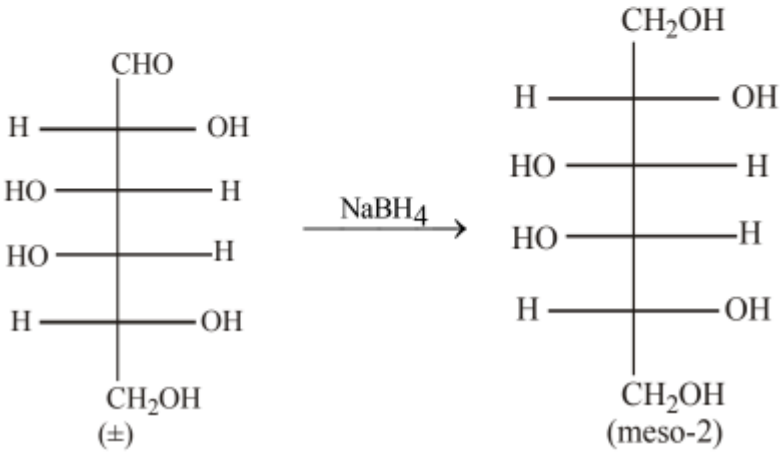


Q.20 Answer:

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Solution:





So, in total 4 stereoisomers are reduced to meso products.

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